

NATASHA THAKUR

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Backend/infrastructure engineer (6+ years) building high-throughput distributed systems. Looking for a role in an environment that provides opportunity to learn and take ownership.

EXPERIENCE

Software Engineer II, Docusign

March 2023 – Present

- Improved customer experience by cutting average API response times by 60% by migrating services to a gRPC-based framework, including a 10M+ record data migration in production with no customer impact.
- Revamped the trial experience landing page by building a new trial management service that handled roughly 15,000 records/hour and 30,000 data-retrieval requests/hour. Independently designed, built, and deployed the service to production, improving homepage customization for new customers.
- Drove ~2,000 leads/day of reliable downstream flow into the sales platform by leading integration of the company's core data-collection architecture with a third-party chatbot.
- Led two major overhauls of the subscription management platform, redesigning the database schema, partitioning, and indexing strategy to support several trial and subscription use cases.
- Reduced churn by 30% by enabling automated upgrade for trial customers complete with retry mechanism and failure management.
- Improved code maintainability and reduced cost by performing an AWS-to-Azure cloud migration.
- Bolstered customers self-serve capabilities by continuously integrating over 10 new offerings into the ecommerce platform.
- Patched 20+ services to meet 7-day security SLAs, built monitoring dashboards (Kafka/Grafana) across 6+ backend services and configured 50+ PagerDuty alerts.

Software Engineer II, Twitter

February 2022 – November 2022

- Drove a 30% increase in throughput by implementing support for "edit tweets" in tools used by content moderation agents.
- Ensured FTC compliance of datasets by implementing appropriate retention and scrubbing policies.
- Integrated backend support for Live Video/Audio within agent tools.

Software Engineer I/II, Nordstrom

March 2020 – February 2022

- Contributed \$3M in ARR by designing and implementing a Node.js microservice to display sponsored ads on the Nordstrom website.
- Built consumers and producers on Nordstrom's Kafka event-streaming platform for real-time event processing.
- Deployed and maintained services on Kubernetes; simplified deployments through CI/CD pipelines and containerization.
- Served as event coordinator for Nordstrom's Women in Tech BRG.

Software Engineering Intern, Bank of America

June 2019 – August 2019

- Delivered 20+ stories during the internship while also fixing defects.
- Contributed to early prototype development of a financial goal-setting and tracking feature within the Bank of America IOS app now used by over 10 million users.
- Coordinated with the design team to implement UI features per specifications.
- Built upon legacy code to integrate new features while ensuring backward compatibility.

EDUCATION

M.S. Computer Science (AI Track), Georgia Institute of Technology *(In Progress)*

B.S. Computer Science, University of Southern California *(December 2019)*

SKILLS & TOOLS

Languages: Python, Java, C/C++, Scala, PHP

Backend & Data: Node.js, Kafka, MySQL, NoSQL, gRPC, Microservices

Cloud & DevOps: AWS, Microsoft Azure, GCP, Kubernetes, Docker, CI/CD

AI/ML: NumPy, scikit-learn, Prompt Engineering, Claude Code, MCP servers, Agentic Workflows

Platforms & Observability: Datadog, Splunk, Grafana, PagerDuty, Linux, Git

AI / ML PROJECTS, GRADUATE COURSEWORK (GEORGIA TECH)

ARC-AGI Reasoning Agent (CS7637): Built a DSL-based agent with a composable solver registry and signature-driven pruning to solve abstract visual-reasoning puzzles.

Gaussian Mixture Models (CS6601): Implemented vectorized K-means and EM-based GMM from scratch in NumPy for image segmentation/compression.

Hidden Markov Models (CS6601): Implemented the Viterbi algorithm and Gaussian emission modeling from scratch for gesture-sequence recognition.